

Data Analytics Case Study: FY2023 Federal Budget Utilization Analysis

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Tools Used: Microsoft Excel, Tableau

1. ASK Phase (The Business Task)

- **Core Goal:** To analyze the spending efficiency and resource allocation of federal entities at the close of Fiscal Year 2023 (Q4).
 - **Objective:** Identify specific spending gaps across agencies—focusing on utilization anomalies—and propose data-driven solutions to optimize fund distribution before the critical September 30th fiscal deadline.
 - **Key Stakeholder Question:** Which sectors are failing to fully utilize their authorized budget authority, what are the primary operational bottlenecks causing these lags, and how can we ensure a target utilization rate of 100%?
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2. PREPARE Phase (Data Sourcing & Integrity)

- **Data Source:** Sourced the public "US Federal Agencies Budget and Spending Data" via Kaggle.
 - **Data Size & Structure:** The dataset consists of 108 records tracking active federal agencies for Fiscal Year 2023, Quarter 4.
 - **Key Variables:** agency_name, outlay_amount (actual spending), obligated_amount (committed funds), and budget_authority_amount (total authorized budget).
 - **Data Integrity:** The data features zero duplicate rows or missing critical financial records, maintaining a high level of credibility and verification for audit purposes.
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3. PROCESS Phase (Data Cleaning & Formatting)

Because the dataset represented trillions of dollars in total government funding, Excel automatically compressed the grand total summaries into complex scientific notation formats (e.g., 1.36153E+13 for total budget authority and microscopic decimals like 8.41622e-07 for individual agency percentage shares).

To ensure the data was readable, accurate, and optimized for visualization, the following cleaning and data enrichment steps were performed in Microsoft Excel:

1. **Metric Engineering (Utilization Rate):** Created a custom calculated column to isolate agency spending efficiency by dividing the actual spending by the authorized budget:

Excel Formula:

= [outlay_amount] / [budget_authority_amount]

2. **Error & Exception Handling:** Anticipated logical system breaks. For inactive accounts or agencies with a budget_authority_amount of \$0, standard division would throw a #DIV/0! error. These anomalies were cleaned and standardized to a clean numeric value of 0.0 (0%), preventing broken data fields upon export.
3. **Data Type Validation:** Verified all core financial columns were explicitly cast as currency/numeric data types with standard decimal precision to avoid aggregation errors in the visualization phase.
4. **Sorting & Prioritization:** Sorted the final worksheet by the newly engineered Utilization Rate in descending order. This immediately pushed critical spending anomalies to the top and severe utilization lags (0%) to the bottom, optimizing the file structure for visual analysis.

4. ANALYZE Phase (Data Insights)

By analyzing the calculated utilization metrics, deep operational insights and significant regional spending variances were discovered:

- **Spending Overages:** Highly active sectors like the Federal Communications Commission (\$127.7%) and the Department of Education (\$104.9%) exceeded their baseline budget authorities, requiring resource shifts. **(Not utilized as “important” in this dataset due to focus being on District Courts only, but is worth mentioning for comparative purposes)**
 - **Significant Spending Gaps:** Across multiple sectors, including specific District Courts, utilization rates plummeted to ranges between **32% and 65%**.
 - **Root Cause Analysis:** Applying domain logic to the numbers reveals that these severe end-of-year funding surpluses are likely driven by hiring lags/vacancies and deferred operational projects that failed to execute before the close of Q4.
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5. SHARE Phase (Data Visualization)

To translate these tabular rows into a narrative stakeholders could act on, the cleaned dataset was imported into Tableau to build an interactive executive dashboard.

- **Dashboard Focus:** The visualizations distinctly highlight the stark contrast between optimized agencies and those suffering from critical spending bottlenecks.
 - **Project Link:** <https://www.linkedin.com/in/madrid-gregory-8b8800358/overlay/1775149900226/single-media-viewer/?profileId=ACoAAFkbpQwBrSi9kgo1HnUc0vdeSg6ptdIRHdU>
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6. ACT Phase (Actionable Recommendations)

To solve the spending gaps and drive the organization toward a **100% utilization rate** by the September 30th end-of-fiscal-year deadline, the following three strategic actions are recommended:

1. **Establish Proactive Quarterly Audits:** Move away from retroactive end-of-year reviews. Implement standardized quarterly internal budget evaluations to catch spending lags in Q1, Q2, and Q3 before they accumulate.
2. **Dynamic Fund Reallocation Framework:** Create an administrative process to dynamically sweep and reallocate projected surplus funds from agencies experiencing hiring delays, redirecting those dollars to over-utilized, high-performing sectors before the fiscal clock runs out.
3. **Project Milestone Tracking:** Tie budget authority to rigid operational project timelines, ensuring that if a project is deferred, the unspent funds are flagged immediately for organizational reuse rather than sitting idle.